GENUS DROPS UNDER BIRATIONAL PROPER MORPHISMS, WITH A RIGIDITY STATEMENT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-curves/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument:

Editorial note. The label lemma-genus-normalization names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Proper birational morphisms are finite in codimension one. Varieties, Lemmas \label{varieties-lemma-finite-in-codim-1} and \label{varieties-lemma-modification-normal-iso-over-codimension-1}. Proof lines 3889-3893: a proper birational morphism f : X' -> X is finite and an isomorphism over a dense open.
- Genus decreases under finite birational maps. Same chapter, \label{lemma-genus-goes-down}. Proof lines 3893-3894: the inequality dim_k H^1(X, O_X) >= dim_k H^1(X', O_{X'}) follows from this lemma.
- Sheaves with zero-dimensional support are generated by global sections. Varieties, Lemma \label{varieties-lemma-chi-tensor-finite}. Proof lines 3901-3904: the cokernel F of O_X -> f_*O_{X'} has zero-dimensional support, hence is generated by global sections, so H^0(X, F) = 0 forces F = 0.

Source reference. Original reference 0CE3: sources/stacks-curves/pinned-original-source.tex, source lines 3850–3850. Prerequisite pointer invoked at proof line(s) 3894; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AYT: sources/stacks-chow/tags.txt, source lines 13801–13801. Prerequisite pointer invoked at proof line(s) 3904; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AB7: sources/stacks-chow/tags.txt, source lines 13010–13010. Prerequisite pointer invoked at proof line(s) 3892; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0BFP: sources/stacks-chow/tags.txt, source lines 14392–14392. Prerequisite pointer invoked at proof line(s) 3893; it names a standard

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fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let k be a field. Let X be a proper scheme of dimension ≤ 1 over k. If $X' \to X$ is a birational proper morphism, then*

$$\dim_k H^1(X, \mathcal{O}_X) \geq \dim_k H^1(X', \mathcal{O}_{X'})$$

If X is reduced, $H^0(X, \mathcal{O}_X) \to H^0(X', \mathcal{O}_{X'})$ is surjective, and equality holds, then $X' = X$.

Proof. If $f : X' \to X$ is proper birational, then it is finite and an isomorphism over a dense open (see Varieties, Lemmas 0AB7 and 0BFP). Thus the inequality by Lemma 0CE3. Assume X is reduced. Then $\mathcal{O}_X \to f_*\mathcal{O}_{X'}$ is injective and we obtain a short exact sequence

$$0 \to \mathcal{O}_X \to f_*\mathcal{O}_{X'} \to \mathcal{F} \to 0$$

Under the assumptions given in the second statement, we conclude from the long exact cohomology sequence that $H^0(X, \mathcal{F}) = 0$. Then $\mathcal{F} = 0$ because $\mathcal{F}$ is generated by global sections (Varieties, Lemma 0AYT). and $\mathcal{O}_X = f_*\mathcal{O}_{X'}$. Since f is affine this implies $X = X'$. □